LINUX ASSIGNMENT

Aditi Biju
261100690027

Part A: Authentication Log

Exercise 1: To print

Write sed commands to:

Q1. Print the first log entry:
sed -n '1p' auth.log

Q2. Print the fifth log entry:

sed -n '5p' auth.log

Q3.Print entries from lines 2 to 6:

sed -n '2,6p' auth.log

Q4.Print the last entry:

sed -n '$p' auth.log

Q5.Print every second log entry starting from line 1:
sed -n '1~2p' auth.log

Q6.Print all entries containing Failed password:

sed -n '/Failed password/p' auth.log

Q7.Print all successful login entries containing Accepted password:

sed -n '/Accepted password/p' auth.log

Q8.Print entries from the first failed login to the authentication failure.:
sed -n '/Failed password/,/authentication failure/p' auth.log

Exercise 2 – Filtering Authentication Events

Q1.Display failed login attempts:
sed -n '/Failed password/p' auth.log

Q2.Display successful login attempts:
sed -n '/Accepted password/p' auth.log

Q3.Display entries containing root:

sed -n '/root/p' auth.log

Q4.Display entries containing Invalid user:

sed -n '/Invalid user/p' auth.log

Q5.Display entries containing the IP address 172.16.0.8.:

sed -n '/172.16.0.8/p' auth.log

Q6.Display all entries except successful logins.:

sed '/Accepted password/d' auth.log

Q7.Display all entries except sudo events.:

sed '/sudo/d' auth.log

Q8.Display entries matching either Failed password or Invalid user.:
sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log

Exercise 3 – Removing Unwanted Entries

Without modifying the original file, display the log after:

Q1.Delete successful login entries:

sed '/Accepted password/d' auth.log

Q2.Delete Connection closed entries:

sed '/Connection closed/d' auth.log

Q3.Delete the first three lines.:

sed '1,3d' auth.log

Q4.Delete lines 4 to 7.:
sed '4,7d' auth.log

Q5.Delete entries containing 192.168.1.15.:

sed '/192.168.1.15/d' auth.log

Q6.Delete every second entry starting from the first.:

sed '1~2d' auth.log

Q7.Display only failed password entries.:

sed '/Failed password/!d' auth.log

Exercise 4 – Classifying Authentication Events

Q1 .Replace Accepted password with [SUCCESSFUL LOGIN].:

sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log

Q2. Replace Failed password with [FAILED LOGIN].:

sed 's/Failed password/[FAILED LOGIN]/g' auth.log

Q3. Replace Invalid user with [INVALID USER].:

sed 's/Invalid user/[INVALID USER]/g' auth.log

Q4. Replace authentication failure with [AUTHENTICATION FAILURE].:

sed 's/authentication failure/[AUTHENTICATION FAILURE]/g' auth.log

Q5. Replace root with PRIVILEGED_USER.:

sed 's/root/PRIVILEGED_USER/g' auth.log

Q6. Replace server with authentication-server.:

sed 's/server/authentication-server/g' auth.log

Exercise 5 – Masking Sensitive Information

Q1. Replace 10.0.0.25 with X.X.X.X.:
sed 's/10.0.0.25/X.X.X.X/g' auth.log

Q2. Replace 172.16.0.8 with SUSPICIOUS_IP.:
sed 's/172.16.0.8/SUSPICIOUS_IP/g' auth.log

Q3. Replace admin with USER_A.:

sed 's/admin/USER_A/g' auth.log

Q4. Replace student with USER_B.:

sed 's/student/USER_B/g' auth.log

Q5. Replace every digit with X.:

sed 's/[0-9]/X/g' auth.log

Q6. Replace periods with hyphens.:
sed 's/\./-/g' auth.log

Q7. Replace all IPv4 addresses with [REDACTED_IP].:

sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log

Q6. Replace periods with hyphens.:
sed 's/\./-/g' auth.log

Q7. Replace all IPv4 addresses with [REDACTED_IP].:

Sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log

PART B – Web-Server Log
Use access.log to solve the below questions

Exercise 6 – Filtering Web Requests

Q1. Display requests with status 200.:
sed -n '/ 200$/p' access.log

Q2. Display requests with status 401.:
sed -n '/ 401$/p' access.log

Q3. Display requests with status 403.:

sed -n '/ 403$/p' access.log

Q4. Display requests to /admin/login.:

sed -n '/\/admin\/login/p' access.log

Q5. Display POST requests.:

sed -n '/POST/p' access.log

Q6. Display requests containing ../.:

sed -n '/\.\.\//p' access.log

Q7. Display requests containing <script>.:

sed -n '/<script>/p' access.log

Q8. Display requests containing OR.:

sed -n '/OR/p' access.log

Q9. Display all requests except successful 200 requests.:

sed '/ 200$/d' access.log

Q10. Display requests from 10.0.0.25.:

sed -n '/^10.0.0.25 /p' access.log

Exercise 7 – Labelling Web Requests

Q1. Replace status 200 with [ALLOWED].:
sed 's/200$/[ALLOWED]/' access.log

Q2. Replace status 401 with [UNAUTHORIZED].:
sed 's/401$/[UNAUTHORIZED]/' access.log

Q3. Replace status 403 with [BLOCKED].:

sed 's/403$/[BLOCKED]/' access.log

Q4.Replace /admin/login with /RESTRICTED_RESOURCE.: sed 's/\/admin\/login/\/RESTRICTED_RESOURCE/g' access.log

Q5. Replace GET with READ_REQUEST.:

sed 's/GET/READ_REQUEST/g' access.log

Q6. Replace POST with SUBMIT_REQUEST.:

sed 's/POST/SUBMIT_REQUEST/g' access.log

Exercise 8 – Grouping Security Operations

Q1. Label and print only 401 requests.:
sed -n '/ 401$/{s/401$/[UNAUTHORIZED]/;p;}' access.log

Q2. Label 403 requests, redact their IP address, and print them.:
sed -En '/ 403$/{s/403$/[BLOCKED]/;s/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/;p;}' access.log

Q3. Replace /admin/login, redact the IP address, and print the matching entries.: sed -En '/\/admin\/login/{s/\/admin\/login/\/PROTECTED_PAGE/g;s/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/;p;}' access.log

Exercise 9 – Adding Incident Labels

Q1. Add SECURITY AUTHENTICATION LOG before the first authentication entry.:

sed '1i SECURITY AUTHENTICATION LOG' auth.log

Q2. Add END OF SECURITY LOG after the final entry.:

sed '$a END OF SECURITY LOG' auth.log

Q3. Add [WARNING] before every failed login.:

sed '/Failed password/i [WARNING]' auth.log

Q4. Add Review this event after every invalid-user entry.:

sed '/Invalid user/a Review this event' auth.log

Q5. Add [SUSPICIOUS WEB REQUEST] before every 403 request.:

sed '/ 403$/i [SUSPICIOUS WEB REQUEST]' access.log

Q6. Add [REQUIRES INVESTIGATION] after every 401 request.:

sed '/ 401$/a [REQUIRES INVESTIGATION]' access.log

Q7. Replace Connection closed lines with [CONNECTION TERMINATED].:

sed 's/.*Connection closed.*/[CONNECTION TERMINATED]/' auth.log

Q8. Replace successful web requests with [NORMAL WEB REQUEST].:

sed '/ 200$/c [NORMAL WEB REQUEST]' access.log

Exercise 10 – Separating Log Entries with G

Q1. Add a blank line after every authentication event.:

sed 'G' auth.log

Q2. Add a blank line after every failed login.:

sed '/Failed password/G' auth.log

Q3. Add a blank line after every blocked web request.:

sed '/ 403$/G' access.log

Q4. Add a blank line after every second authentication entry starting from line 1.:

sed '1~2G' auth.log

Q5. Write failed logins to failed_logins.txt.:

sed -n '/Failed password/w failed_logins.txt' auth.log

Q6. Write successful logins to successful_logins.txt.:

sed -n '/Accepted password/w successful_logins.txt' auth.log

Q7. Write invalid users to invalid_users.txt.:

sed -n '/Invalid user/w invalid_users.txt' auth.log

Q8. Write unauthorized requests to unauthorized_requests.txt.:

sed -n '/ 401$/w unauthorized_requests.txt' access.log

Q9. Write blocked requests to blocked_requests.txt.:

sed -n '/ 403$/w blocked_requests.txt' access.log

Q10. Write entries containing 172.16.0.8 to suspicious_ip.txt.:

sed -n '/172.16.0.8/w suspicious_ip.txt' auth.log

Exercise 11 – Combined sed Operations

Q1. Label successful and failed authentication events in one command.:

sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' -e 's/Failed password/[FAILED LOGIN]/g' auth.log

Q2. Label 401 and 403 web responses in one command.:

sed -e 's/401$/[UNAUTHORIZED]/' -e 's/403$/[BLOCKED]/' access.log

Q3. Delete successful logins and print only failed logins.:

sed -n '/Accepted password/d;/Failed password/p' auth.log

Q4. Replace root with [PRIVILEGED_USER] and redact IP addresses.:

```
sed -E 's/root/[PRIVILEGED_USER]/g; s/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
```

Q5. Replace GET and POST with readable request labels.:

```
sed -e 's/GET/READ_REQUEST/g' -e 's/POST/SUBMIT_REQUEST/g' access.log
```

Exercise 13 – Creating security_report.sed

```
cat > security_report.sed <<'EOF'
1i SECURITY INCIDENT REPORT
s/Failed password/[FAILED LOGIN]/g
s/Invalid user/[INVALID USER ATTEMPT]/g
s/Accepted password/[SUCCESSFUL LOGIN]/g
s/[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/g
$a END OF INCIDENT REPORT
EOF
```

Q1. Run the sed script on auth.log.:

```
sed -f security_report.sed auth.log
```

Q2. Save the generated report to incident_report.txt.:

```
sed -f security_report.sed auth.log > incident_report.txt
```

Exercise 14 – Modifying a Security Configuration File

```
cat > security.conf <<EOF
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
LogLevel INFO
GuestLogin enabled
EOF
```

Q1. Create a backup before editing.:

```
cp security.conf security.conf.backup
```

Q2. Disable root login.:

sed -i 's/PermitRootLogin yes/PermitRootLogin no/' security.conf

Q3. Reduce MaxAuthTries from 6 to 3.:

sed -i 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf

Q4. Enable the firewall.:

sed -i 's/FirewallStatus disabled/FirewallStatus enabled/' security.conf

Q5. Change the log level to VERBOSE.:

sed -i 's/LogLevel INFO/LogLevel VERBOSE/' security.conf

Q6. Disable guest login.:

sed -i 's/GuestLogin enabled/GuestLogin disabled/' security.conf Expected final contents:

```
PermitRootLogin no
PasswordAuthentication yes
MaxAuthTries 3
FirewallStatus enabled
LogLevel VERBOSE
GuestLogin disabled
```

Bash Script – security_log_analyser.sh

```
#!/bin/bash

if [ $# -eq 0 ]; then
   echo "Usage: $0 <logfile>"
   exit 1
fi

if [ ! -f "$1" ]; then

   echo "Error: File does not exist"
   exit 1
fi

echo "SECURITY ALERT REPORT" > security_alert_report.txt

sed -n '/Failed password/p' "$1" >> security_alert_report.txt
```

```
sed -n '/Invalid user/p' "$1" >> security_alert_report.txt

sed -i -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' security_alert_report.txt

sed -i 's/^/[ALERT] /' security_alert_report.txt

echo "END OF REPORT" >> security_alert_report.txt

cat security_alert_report.txt
```

Q1. Make the script executable.:

```
chmod +x security_log_analyser.sh
```

Q2. Run the script using auth.log.:

```
./security_log_analyser.sh auth.log
```